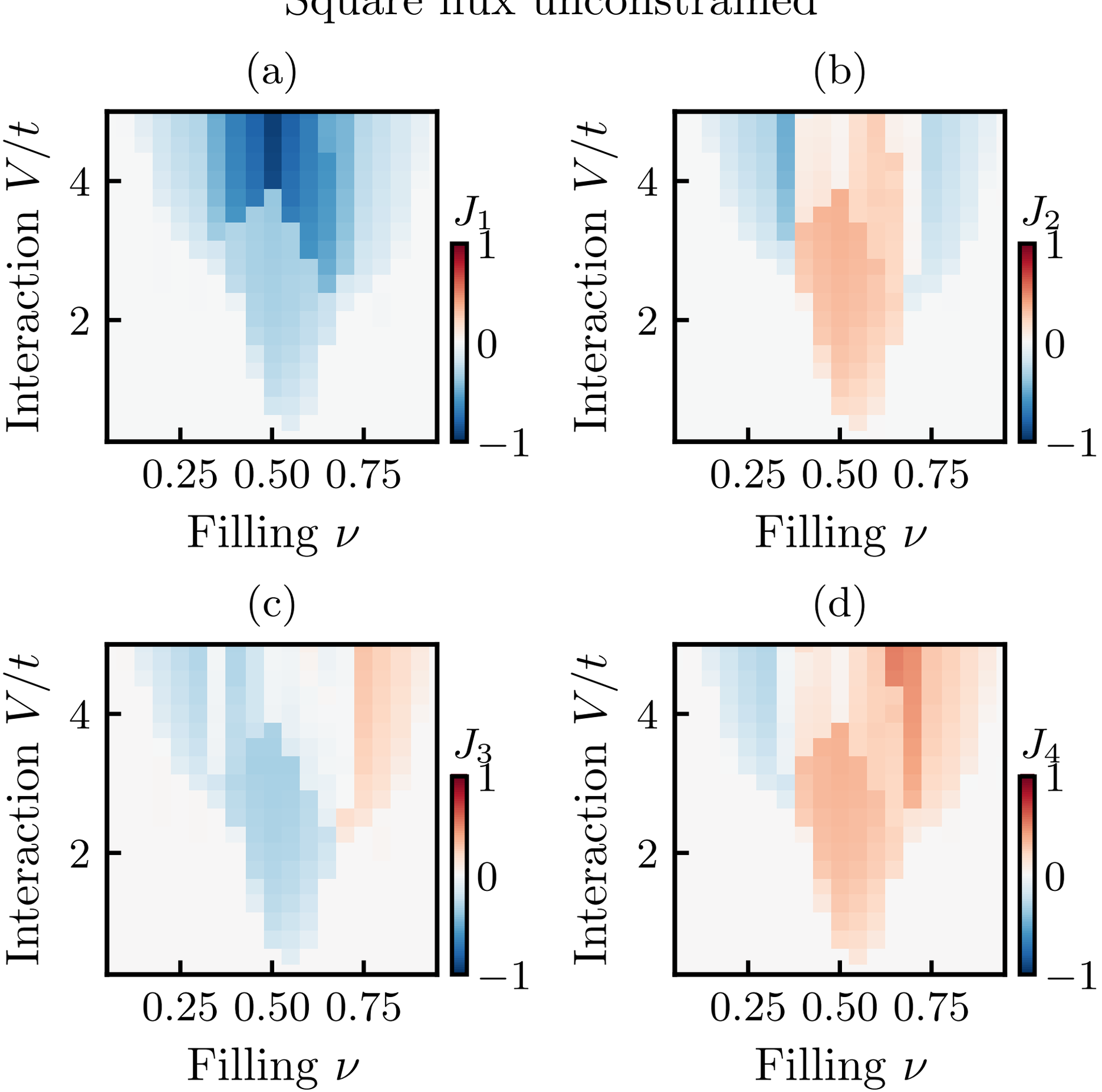

Square flux unconstrained
(a)
(b)
(c)
(d)
Interaction V/t
Filling ν
J_1
J_2
J_3
J_4
2
4
0.25 0.50 0.75
1
0
−1